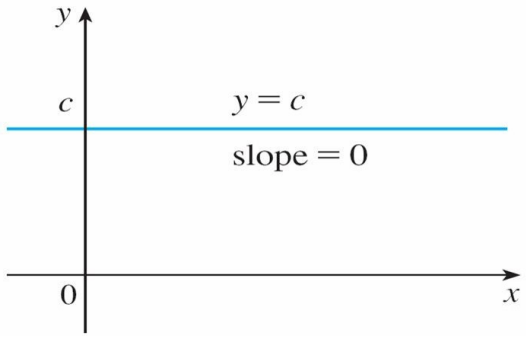
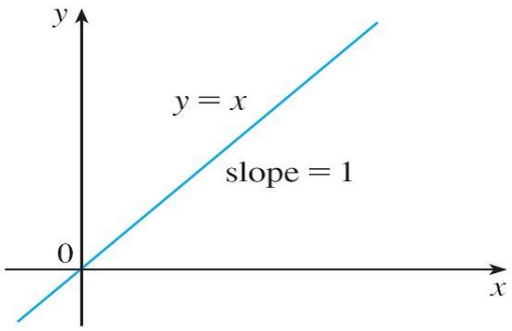
CHAPTER 3 DIFFERENTIAL RULES

3.1 Derivatives of Polynomials and Exponential Functions

- **Rules for Finding Derivatives**

- $\dfrac{d}{dx}c = 0$
- $\dfrac{d}{dx}x = 1$
- $\dfrac{d}{dx}x^n = nx^{n-1}$, n is a positive integer.





- For any real number n, $\dfrac{d}{dx}x^n = nx^{n-1}$.

- $D_x\left[kf(x)\right] = kD_xf(x)$
- $D_x\left[f(x) \pm g(x)\right] = D_xf(x) \pm D_xg(x)$

Example 3.1. Find the derivatives of the following functions:

(a) $y = 5x^2 + 7x - 6$

(b) $y = 4x^6 - 3x^5 - 10x^2 + 5x + 16$

(c) $y = \dfrac{x^2 + 1}{x}$

(d) $y = \dfrac{x + 1}{\sqrt{x}}$

Example 3.2. Find the points on the curve $y = x^4 - 6x^2 + 4$ where the tangent line is horizontal.

Example 3.3. The equation of motion of a particle is $s = 2t^3 - 5t^2 + 3t + 4$, where s is measured in centimeters and t in seconds. Find the acceleration as a function of time. What is the acceleration after 2 seconds?

- **Derivatives of Exponential Functions**

- **The Definition of e**

e is the number such that $\lim_{h\to 0} \dfrac{e^h - 1}{h} = 1.$

- **Derivative of Natural Exponential Function**

$$\frac{d}{dx}e^x = e^x$$

Example 3.4. At what point on the curve $y = e^x$ is the tangent line parallel to the line $y = 2x$?

Exercises 3.1 3, 5, 7, 11, 13, 17, 19, 21, 23, 25, 29, 31, 33, 35, 37, 49, 55, 57, 59, 65, 75.

3.2 Product and Quotient Rules

Example 3.5. Given $g(x) = x$, $h(x) = x^2$, and $f(x) = g(x)h(x)$, find $D_x g(x)$, $D_x h(x)$, $D_x f(x)$, and show that $D_x[g(x)h(x)] \neq [D_x g(x)]\,[D_x h(x)]$.

- **Product Rule**

If u and v are differentiable functions of x,

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

Example 3.6. Find the derivative of $y = (3x^2-5)(2x^4-x)$ by using the product rule. Check the answer by first expanding the product and then take derivative.

Example 3.7. Given $f(x) = xe^x$,

(a) Find $f'(x)$.

(b) Find $f^{(n)}(x)$.

Example 3.8. Given $f(t) = \sqrt{t}(a + bt)$, find $f'(t)$.

Example 3.9. Given $f(x) = \sqrt{x}g(x)$, where $g(4) = 2$, $g'(4) = 3$, find $f'(4)$.

- **Quotient Rule**

If u and v are differentiable functions of x,

$$\frac{d}{dx}\frac{u}{v} = \frac{v\dfrac{du}{dx} - u\dfrac{dv}{dx}}{v^2}$$

at the point where $v(x) \neq 0$.

Example 3.10. Find $\dfrac{d}{dx}\left(\dfrac{3x-5}{x^2+7}\right)$.

Example 3.11. Find the equation of the tangent line to the curve $y = \dfrac{e^x}{1+x^2}$ at the point $\left(1, \dfrac{e}{2}\right)$.

Exercises 3.2 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 31, 43, 45, 47, 49, 55.

3.3 Derivatives of Trigonometric Functions

- **Special Trigonometric Limits**

(a) $\lim_{t\to 0} \dfrac{\sin t}{t} = 1$

(b) $\lim_{t\to 0} \dfrac{1-\cos t}{t} = 0.$

- **Theorem**

- $\dfrac{d}{dx}\sin x = \cos x$
- $\dfrac{d}{dx}\cos x = -\sin x$

Example 3.12. Find $\dfrac{d}{dx}(3\sin x - 2\cos x)$.

Example 3.13. Find the equation of the tangent line to the curve $y = 3\sin x$ at the point $(\pi, 0)$.

Example 3.14. Find $\dfrac{d}{dx}\left(x^2 \sin x\right)$.

Example 3.15. Find $\dfrac{d}{dx}\left(\dfrac{1+\sin x}{\cos x}\right)$.

- **Theorem**

  - $\dfrac{d}{dx}\tan x = \sec^2 x$
  - $\dfrac{d}{dx}\cot x = -\csc^2 x$
  - $\dfrac{d}{dx}\sec x = \sec x \tan x$
  - $\dfrac{d}{dx}\csc x = -\csc x \cot x$

Example 3.16. Find the equation of the tangent line to the curve $y = \tan x$ at the point $\left(\frac{\pi}{4}, 1\right)$.

Example 3.17. Find $\frac{d}{dx}\left(x^n \cot x\right)$ where n is an integer.

Example 3.18. Find all the points on the graph of $y = \sin^2 x$ where the tangent line is horizontal.

Exercises 3.3 1, 3, 5, 7, 9, 11, 13, 15, 17, 18, 19, 20, 23, 33, 35, 39, 41, 43, 45, 51, 57 (*Hint*: $s = r\theta$, draw the bisector of the angle θ and find $\sin(\frac{\theta}{2})$).

3.4 The Chain Rule

Example 3.19. Find $\dfrac{d}{dx}(x^2+2x)^3$.

- **Chain Rule**

> Let $y=f(u)$ and $u=g(x)$. If g is differentiable at x and f is differentiable at $u=g(x)$, then the composite function $f\circ g$, defined by $(f\circ g)(x)=f(g(x))$, is differentiable at x and
>
> $$(f\circ g)'(x)=f'(g(x))\,g'(x)$$
>
> or
>
> $$\frac{dy}{dx}=\frac{dy}{du}\frac{du}{dx}$$

Example 3.20. If $y = \left(2x^2 - 4x + 1\right)^{60}$, find $D_x y$.

Example 3.21. If $y = \dfrac{1}{\sqrt[3]{2x^5 - 7}}$, find $\dfrac{dy}{dx}$.

Example 3.22. Find $D_t \left(\dfrac{t^3 - 2t + 1}{t^4 + 3}\right)^{13}$.

Example 3.23. If $y = \sin 2x$, find $\dfrac{dy}{dx}$.

Example 3.24. Find $F'(y)$ if $F(y) = y \sin(y^2)$.

Example 3.25. Find $D_x \dfrac{x^2(1-x)^3}{1+x}$.

Example 3.26. Find $\dfrac{d}{dx}e^{\sin x}$.

Example 3.27. Find $\dfrac{d}{dx}a^x$, where $a > 0$, $a \neq 1$.

Example 3.28. Find $\dfrac{d}{d\theta}2^{\sec\theta}$.

Example 3.29. If F is differentiable, find

(a) $D_x\left[F(x^3)\right]$

(b) $D_x\left[(F(x))^3\right]$.

Example 3.30. Find $D_x \sin^3(4x)$.

Example 3.31. Find $D_x \sin\left[\cos\left(x^2\right)\right]$.

Exercises 3.4 7, 9, 11, 13, 15, 17, 25, 27, 29, 31, 43, 51, 53, 61, 71, 73, 79, 81.

3.5 Implicit Differentiation

- The relation $y^3+7y = x^3$ can define y as a function of x. It is called an **implicit function**.

  Example 3.32. Find $\dfrac{dy}{dx}$ if $y^3 + 7y = x^3$.

Example 3.33. Find the equation of the tangent line to the curve $4x^2y - 3y = x^3$ at the point $(1, 1)$.

Example 3.34. At what point in the first quadrant is the tangent line to the curve $x^3 + y^3 = 6xy$ horizontal?

Example 3.35. Given $\sin(x+y) = y^2 \cos x$, find $\dfrac{dy}{dx}$.

Example 3.36. Given $x^4 + y^4 = 16$, find y''.

- **Derivatives of Inverse Trigonometric Functions**

  - $D_x \sin^{-1} x = \dfrac{1}{\sqrt{1-x^2}}$
  - $D_x \cos^{-1} x = -\dfrac{1}{\sqrt{1-x^2}}$
  - $D_x \tan^{-1} x = \dfrac{1}{1+x^2}$

Example 3.37. Differentiate $y = \dfrac{1}{\sin^{-1} x}$.

Example 3.38. Differentiate $y = x \arctan \sqrt{x}$.

Exercises 3.5 5, 7, 9, 11, 13, 15, 17, 19, 27, 35, 49, 51, 57, 63, 75.

3.6 Derivatives of Logarithmic Functions

- **Derivatives of Logarithmic Functions**

- $D_x \ln x = \dfrac{1}{x}$
- $D_x \log_a x = \dfrac{1}{x \ln a}$

Example 3.39. Find $D_x \ln \sqrt{x}$.

Example 3.40. Find $D_x \ln\left(x^2 - x - 2\right)$.

Example 3.41. Show that

$$D_x \ln|x| = \frac{1}{x}$$

Example 3.42. Find $\dfrac{d}{dx}\ln(1+\cos x)$.

Example 3.43. Find $\dfrac{d}{dx}\sqrt{\ln x}$.

Example 3.44. Find $\dfrac{dy}{dx}$ if $y = \ln\sqrt[3]{\dfrac{x-1}{x^2}}$, $x > 1$.

Example 3.45. Differentiate $y = \dfrac{\sqrt{1-x^2}}{(x+1)^{\frac{2}{3}}}$.

Example 3.46. Differentiate $y = x^x$.

Example 3.47. Differentiate $y = \left(x^2 + 1\right)^{\sin x}$.

- **e as a Limit**

$$\lim_{x\to 0}(1+x)^{\frac{1}{x}} = e$$
$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n} = e$$

Example 3.48. Find $\lim_{n\to\infty}\left(\frac{n+2}{n}\right)^{n}$.

Example 3.49. Find $\lim_{n\to\infty}\left(\frac{n-2}{n}\right)^{n}$.

Exercises 3.6 3, 5, 7, 9, 13, 15, 19, 33, 39, 41, 47, 56.

3.7 Rates of Change

Example 3.50. The position of a particle is given by

$$s = f(t) = t^3 - 6t^2 + 9t$$

where s is measured in meters and $t \geq 0$ is measured in seconds.

(a) Find the velocity at time t.

(b) What is the velocity after 2 s? After 4 s?

(c) When is the particle at rest?

(d) When is the particle moving to the right?

(e) Find the total distance travelled by the particle in the first 5 seconds.

(f) Find the acceleration at time t.

(g) When is the particle speeding up? When is it slowing down?

Exercises 3.7 1(a)(b)(c)(d)(e)(g)(i).

3.9 Related Rates

- If a variable y depends on time t, then its derivative $\dfrac{dy}{dt}$ is called a time rate of change.

Example 3.51. Air is pumped into a spherical balloon so that its volume increases at a rate of 100 cm^3/s. How fast is the radius of the balloon increasing when the diameter is 50 cm?

Example 3.52. A small balloon is released at a point 150 m away from an observer, who is on level ground. If the balloon goes straight up at a rate of 8 m/s, how fast is the distance from the observer to the balloon increasing when the balloon is 50 m high?

Example 3.53. Water is pouring into a conical tank at the rate of 8 m^3 per second. If the height of the tank is 12 m and the radius of its circular opening is 6 m, how fast is the water level rising when the water is 4 m deep?

Exercises 3.9 3, 5, 17, 22 (1.01 m/s), 25, 32 (-0.25 rad/s), 45.

3.10 Linear Approximations and Differentials

- $$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

  When $x - a$ is small, $f'(a) \approx \dfrac{f(x) - f(a)}{x - a}$, and yields

  $$f(x) - f(a) \approx (x - a) f'(a)$$

  $$f(x) \approx f(a) + f'(a)(x - a)$$

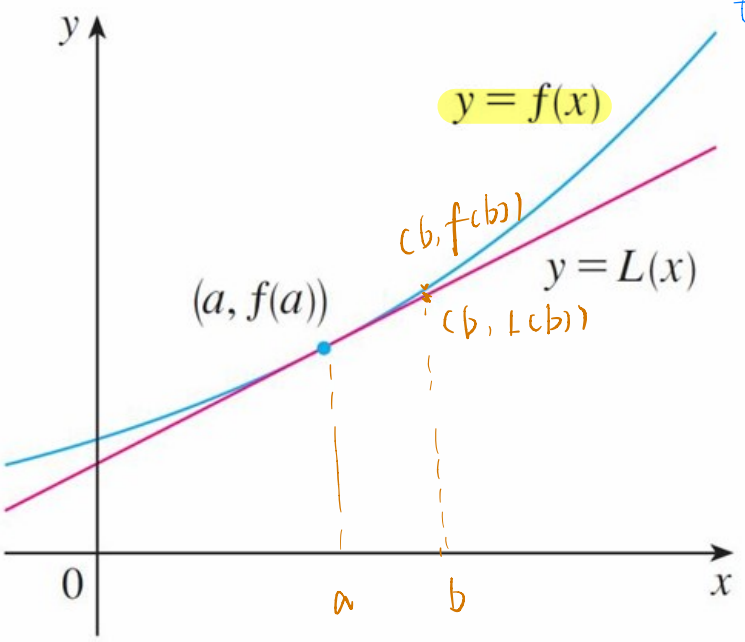
- **Linearization**

  tangent line 的对于

  $$L(x) = f(a) + f'(a)(x - a)$$

  is called the **linearization** of f at a.

  用 curve 上的点 approximate tangent line 上的点



Example 3.54. Find approximations for $\sqrt{4.6}$ and $\sqrt{8.2}$.

let $f(x) = \sqrt{x}$

$f'(x) = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$

$\sqrt{1} = 1$

$\sqrt{4} = 2$

$\sqrt{9} = 3$.

$L(x) = f(a) + f'(a)(x-a)$

$\sqrt{4.6} = f(4.6) \approx L(4.6) = f(4) + f'(4)(4.6-4)$

$= 2 + \frac{1}{4} \times 0.6$

$= 2.15$

remark: $\sqrt{4.6} = 2.1448$

a很接近x且好算

$L(8.2) = f(9) + f'(9)(8.2-9)$

$= \sqrt{9} + \frac{1}{2\sqrt{9}}(8.2-9)$

$= 3 + \frac{1}{6} \times 0.8$

$= 2.8667$

$L(x)$

$dy = \sqrt{9} - 2.8667$

$f(x)$

$\Delta y = \sqrt{9} - 2.8636$

remark: $\sqrt{8.2} = 2.8636$

- **Differentials**

Let $y = f(x)$ be a differentiable function of the independent variable x.

- Δx is an arbitrary increment in the independent variable x.
- dx, called the **differential** of the independent variable x, is equal to Δx.
- Δy is the actual change in the variable y as x changes from x to $x + \Delta x$; that is

$$\Delta y = f(x + \Delta x) - f(x)$$

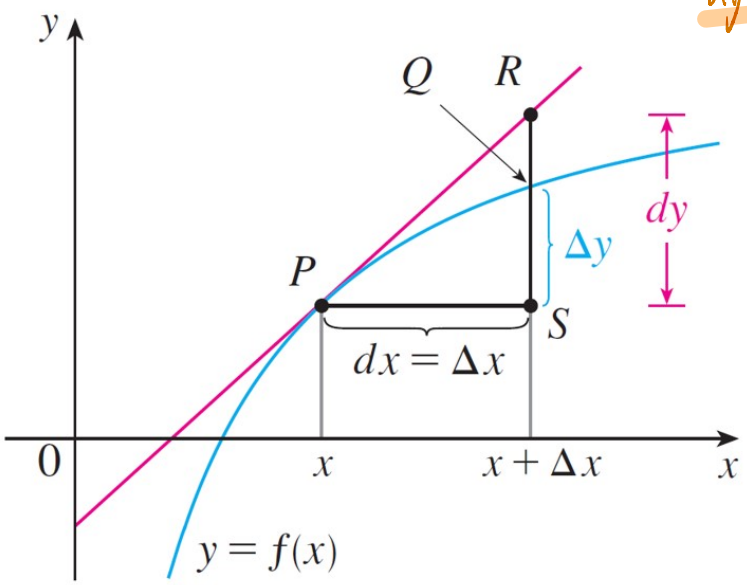
$f(x) \approx f(a) + f'(a)(x-a) \Rightarrow f(x) - f(a) = f'(a)(x-a)$
let $f(x) - f(a) = \Delta y$
$x - a = \Delta x$

- dy, called the **differential** of the dependent variable y, is defined as

$$dy = f'(x)dx$$

$f'(x) = \frac{dy}{dx}$

$\Delta y = f'(a)\Delta x$
let $dx = \Delta x$
$\Delta y = f'(a)dx$
$dy \approx f'(a)dx$



Example 3.55. Compute the value of Δy and dy if $f(x) = x^3 + x^2 - 2x + 1$ and x changes from 2 to 2.05.

actual difference

approximate difference.

$$\Delta y = f(2.05) - f(2)$$

$$= 2.05^3 + 2.05^2 - 2 \times 2.05 + 1 - (2^3 + 2^2 - 2 \times 2 + 1)$$

$$= 0.717625$$

$$dy = f'(x)\,dx$$

$$f'(x) = 3x^2 + 2x - 2$$

$$dy = (3x^2 + 2x - 2)\,dx$$

when $x = 2$ and $dx = 2.05 - 2 = 0.05$

$$dy = 14 \times 0.05$$

$$= 0.7$$

Example 3.56. Approximate the increase in the area of a soap bubble when its radius increases from 3 cm to 3.025 cm.

let V be the volume of the bubble and r be the radius

$V = \frac{4}{3}\pi r^3$ $\quad V'(r) = 4\pi r^2$

$dV = V'(r)\,dr$

$dV = \frac{4}{3}\pi \times 3r^2 \times \frac{dr}{dt} \times$ 变化率 rate of change 半径

$dV = 4\pi r^2\,dr$

when $r = 3$ $\quad dr = 0.025$

$dV = 4\pi \times 3^2 \times 0.025$

Example 3.57. The side of a cube is measured as 11.4 cm with a possible error of ± 0.05 cm. Evaluate the volume of the cube and give an estimate for the possible error in this value.

let V be the volume of cube

x be the length

$V = x^3$

$\frac{dV}{dx} = 3x^2$

$dV = 3x^2 \cdot dx$

$= 3 \times (11.4)^2 \times (0.05)$

$= 19.494$

$V = x^3 = (11.4)^3 = 1481.544$

So the volume of the cube is $1481.544\,cm^3 \pm 19.494\,cm^3$

Example 3.58. Poiseuille's Law for blood flow says that the volume flowing through an artery is proportional to the fourth power of the radius, that is, $V = kR^4$. By how much must the radius be increased in order to increase the blood flow by 50%?

$V = kR^4$

$\frac{dV}{dr} = 4kR^3$

$dV = 4kR^3 \cdot dr$

$dV = \frac{dV}{dr} \cdot dR$

$\frac{dV}{V} = \frac{\frac{dV}{dr} \cdot dR}{V}$

$0.5 = \frac{4kR^3 \cdot dr}{V}$

$0.5 = \frac{4kR^3 dr}{kR^4}$

$0.5 = 4 \cdot \frac{dR}{R}$

$dR = \frac{0.5R}{4}$

$= \frac{R}{8}$

$\frac{dV}{V} = 0.5$

$\frac{dV}{V} = \frac{4kR^3 \cdot dr}{V}$

$0.5V = 4kR^3 dr$

$dr = \frac{0.5V}{4kR^3}$

$dr = \frac{0.5 \cdot kR^4}{4kR^3}$

$= \frac{0.5R}{4}$

$= \frac{R}{8}$

Exercises 3.10 19, 25, 33, 35, 40 (20%).